



Vitamin K Deficiency: Diagnosis and Management

Natalie Mathews , M.D., C.M.^{1,2,3} and Catherine P. M. Hayward , M.D., Ph.D.^{4,5,6}

¹Division of Hematology-Oncology, Department of Pediatrics, Centre Hospitalier Universitaire Sainte-Justine, Université de Montréal, Canada; ²Department of Clinical Laboratory Medicine, OPTILAB Montréal-CHU Sainte-Justine, Université de Montréal, Canada; ³Centre Hospitalier Universitaire Sainte-Justine Azrieli Research Center, Montreal, Quebec, Canada; ⁴Department of Pathology and Molecular Medicine, McMaster University, Hamilton, Canada; ⁵Hamilton Regional Laboratory Medicine Program, Hamilton, Canada; ⁶Department of Medicine, McMaster University, Hamilton, Canada

Vitamin K (VK) deficiency (VKD) commonly causes coagulopathy across the age spectrum. The reduced form of VK is an essential cofactor for the post-translational γ -carboxylation of coagulation factors (Fs) II, VII, IX, and X; proteins C and S; and additional proteins. This carboxylation creates high-affinity calcium-binding sites that are important for their functions. VK is a fat-soluble vitamin, with half of the daily needs met by vitamin K1 from the diet (particularly green leafy vegetables) and the other half met by vitamin K2 produced by gut flora. VKD can develop within days because of limited tissue stores of VK. VKD increases risks for bleeding, with neonates and infants at the highest risk unless they are administered routine VK prophylaxis at birth. Diagnosing VKD is challenging because of the different forms and half-lives of VK isoforms. Often, patients with suspected VKD-related coagulopathies are assessed for coagulopathy correction within 1–2 days after receiving VK, either orally or parenterally. VKD increases the plasma levels of proteins induced by the absence of VK, such as des- γ -carboxylated factor II, which is more commonly used as a biomarker for hepatocellular carcinoma than as a VKD biomarker. VKD causes notable discrepancies in FII levels measured by factor assays based on the prothrombin time (PT) rather than Ecarin reagents, as the Ecarin prothrombin activator directly converts normal and des- γ -carboxylated FII to meizothrombin (resulting in higher FII levels than estimated with PT reagents). In this review, we summarize current information on the causes, consequences, diagnosis, prevention, and treatment of VKD.

Key Words: Blood coagulation disorders, Blood coagulation factor deficiencies, Blood coagulation tests, Vitamin K, Vitamin K deficiency

Received: October 29, 2024

Revision received: January 2, 2025

Accepted: March 1, 2025

Published online: April 24, 2025

Corresponding author:

Catherine P. M. Hayward, M.D., Ph.D.,
F.R.C.P.(C)

Departments of Pathology and Molecular
Medicine, McMaster University and the
Hamilton Regional Laboratory Medicine
Program, Hamilton, ON, Canada, Room
2N29A, McMaster University Medical
Centre, 1200 Main St. West, Hamilton,
Ontario L8N 3Z5, Canada

E-mail: haywrdc@mcmaster.ca



© Korean Society for Laboratory Medicine

This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<https://creativecommons.org/licenses/by-nc/4.0>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

INTRODUCTION

Vitamin K (VK) deficiency (VKD) is a very common acquired coagulopathy that can cause severe bleeding [1–3]. VK is a fat-soluble vitamin that is essential for the post-translational γ -carboxylation of coagulation factors (Fs) II, VII, IX, and X; the anticoagulant proteins C, S, and Z; and several extrahepatic proteins that include osteocalcin [2, 3], matrix Gla protein, and Gla-rich protein, which are involved in bone, matrix, and vasculature homeostasis. γ -carboxylation of VK-dependent proteins is impor-

tant for their functions, and it creates high-affinity binding sites for calcium that mediate phospholipid binding [4, 5]. VKD reduces the functional levels of procoagulant and anticoagulant proteins and typically presents with bleeding or asymptomatic coagulation-test abnormalities [2, 6]. Although most causes of VKD are acquired, inherited defects represent rare but important causes of VKD that are not corrected with VK replacement (reviewed in [7]).

Physiologically, approximately half of the daily VK requirement comes from the diet, and the other half is produced by the gut

flora [3]. Unlike other fat-soluble vitamins, VK is minimally stored in tissues [8], and VKD can develop rapidly within days of reduced dietary intake or absorption. Although VKD affects people of all ages [2], neonates are particularly vulnerable to developing vitamin K deficiency bleeding (VKDB) primarily because of low placental transfer of VK [9].

FORMS AND FUNCTIONS OF VK

VK exists in two main isoforms that share a 2-methyl-1,4-naphthoquinone structure, namely the phyloquinone (vitamin K1, VK1) and menaquinone forms (vitamin K2, VK2) [3, 10]. Menaquinones have both short- and long-chain forms, with longer half-lives than VK1 [3]. Most VK found within food is VK1 [3]. Green leafy vegetables, particularly kale, parsley, collard greens, and spinach, are particularly abundant sources of VK1 [1, 11]. VK2 is synthesized by gut flora, with nutritional sources (meat, dairy products, and fermented foods) representing a minor source of VK2 [1, 3].

Fig. 1 summarizes the function of VK and the key steps involved in the γ -carboxylation of VK-dependent proteins [4]. The

reduced form of VK is an essential cofactor for γ -glutamyl carboxylase (GGCX), which converts Glu residues in VK-dependent proteins to γ -carboxyglutamic acid (Gla domains) [4]. Reduced VK is then regenerated by vitamin K epoxide reductase (VKOR). Warfarin and other VK antagonists and rodenticides cause VKD by inhibiting VKOR activity [4, 5, 8, 12].

RISK FACTORS FOR VKD AND VKDB

VKD is a very common coagulopathy among inpatients and outpatients referred for coagulation-test abnormalities, with infants and neonates being at particularly higher risk for VKDB [1, 6, 11]. Table 1 summarizes the clinical symptoms of congenital and acquired VKD, which predominantly involve bleeding, except for asymptomatic cases that are identified based on coagulation abnormalities.

The first description of VKDB in neonates and infants dates back to 1894 when Dr. Charles Townsend described a hemorrhagic and sometimes fatal disease in newborns [13] that is now called VKDB. The discovery of VK in 1935 [14] was pivotal to understanding the cause of VKDB. VKDB affects ~0.6% of

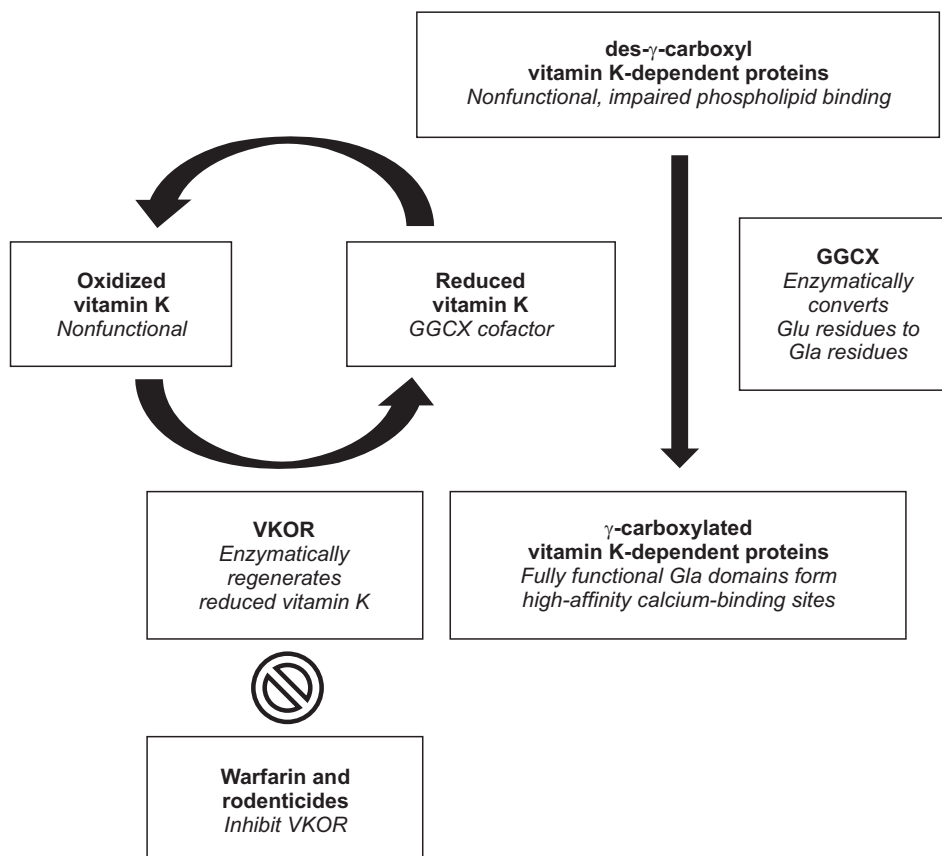


Fig. 1. Key enzymatic pathways involved in the post-translational γ -carboxylation of vitamin K-dependent proteins. Abbreviations: GGCX, γ -glutamyl carboxylase; Gla, γ -carboxyglutamic acid; VKOR, vitamin K epoxide reductase.

Table 1. Clinical manifestations of vitamin K deficiency by age

Age	Symptoms of bleeding from vitamin K deficiency
Prenatally or first 24 hrs of life	ICH, cephalohematoma, intraabdominal bleeding
Days 2–7 of life	Bruising, umbilical bleeding, GI bleeding, phlebotomy bleeding, ICH
Day 8–6 months of life	ICH (majority), skin bleeding, epistaxis, GI bleeding, umbilical bleeding, surgical bleeding
Children >6 months and adults	Relatively few reports. It can include GI bleeding, challenge-related bleeding (e.g., surgical-site bleeding), and rarely ICH. Often asymptomatic with laboratory abnormalities suggesting the possibility of vitamin K deficiency

Abbreviations: GI, gastrointestinal; ICH, intracranial hemorrhage.

newborns and typically presents within a few days of birth [1, 15]. The reason that neonates are at high risk for VKDB is because they have low levels of VK-dependent proteins caused by inefficient placental transfer of VK, low VK production by gut flora, and low gastrointestinal VK absorption [1, 11, 16]. These features result in low hepatic stores of VK at birth, low VK levels in cord blood, and elevated plasma levels of protein induced by vitamin K absence (PIVKA)-II (> 10 ng/mL), a hypo- γ -carboxylated form of FII that is produced under VK deficiency [1, 11, 15, 16].

VKDB in infancy is subclassified according to the timing of onset as early (≤ 24 hrs of life), classic (within the first week of life), or late (within the first 6 months of life) [1]. Early-onset VKDB affects ~6–12% of at-risk newborns not administered VK supplementation [11] and can manifest with intracranial bleeding (in approximately 20–25% of cases), cephalohematoma, and/or intraabdominal hemorrhages [11, 17]. Risks for early VKDB are increased by maternal medications that interfere with VK metabolism, such as VK antagonists (warfarin and coumarin), anti-epileptics (phenytoin, carbamazepine, and barbiturates), certain antibiotics (cephalosporins), and antituberculosis drugs (isoniazid and rifampicin) [11, 16]. Maternal hyperemesis gravidarum, eating disorders, and bariatric surgery can cause life-threatening early VKDB in neonates [18–21]. Notably, some mothers with these risk factors showed reduced VK1 or PIVKA levels despite normal coagulation parameters [19, 22], suggesting that diagnosing VKD may be particularly challenging within this population. Nonetheless, at-risk fetuses may develop antenatal intracranial hemorrhages, and early-risk recognition in pregnant mothers and prenatal VK administration may be the only way to prevent such severe complications [18, 23].

Classic VKDB is associated with illness and/or poor feeding and affects ~1.5% of untreated infants [16]. Classic VKDB is more prevalent among breastfed infants because of the low VK content of breast milk (~2.5 mg VK/L vs. 24–175 mg/L for formula feeds) [1, 15]. Classic VKDB tends to present with bruising, bleeding from venipuncture sites, umbilical bleeding, gastro-

intestinal bleeding, and, on rare occasions, intracranial hemorrhage [11, 17].

Late VKDB is rare, except in regions of the world where prophylactic VK at birth is not a routine practice [16, 24]. VK prophylaxis at birth reduces the incidence of late VKDB from 10–80 per 100,000 births down to 0.24–3.2 per 100,000 births [24]. Late VKDB is often considered the most severe form of VKDB [17], as the mortality is ~10–20% [11, 16, 17], and ~50–75% of cases present with an acute, intracranial hemorrhage (ICH) with long-term neurological sequelae in ~20–50% of affected individuals [11, 16, 17]. Late VKDB can also manifest with cutaneous, nasal, gastrointestinal, umbilical, and/or surgical bleeding [16, 24]. In approximately 1/3 of cases of late VKDB, ecchymoses precede an ICH by several weeks [17]. Although late VKDB is almost exclusive to breastfed infants [11], additional risk factors for VKD typically include cholestatic liver disease (e.g., from biliary atresia or alpha-1 antitrypsin deficiency), malabsorption, or recent antibiotic therapy [24]. VKDB may be the first sign that an infant has one of these underlying disorders [1].

In older children and adults, VKD typically reflects poor intake, malabsorption, hepatic disorders, or antibiotic therapy. The results of a recent cohort study showed that common comorbidities among patients evaluated for VKD included a history of bleeding and factors such as recent antibiotic therapy, chronic kidney disease, treatment or exposure to a vitamin K antagonist (VKA), gastrointestinal diseases, and bowel surgery [6]. However, none of these factors were strongly predictive of VKD [6]. Often, VKD is subclinical, manifesting as an elevated prothrombin time (PT)/international normalized ratio (INR) that may trigger other coagulopathy investigations. Nonetheless, serious and sometimes life-threatening bleeding can occur with VKD [2].

The risk of bleeding caused by VKD is particularly high for people with rodenticide poisoning from long-acting VKAs, such as brodifacoum. Such poisoning may reflect intentional or accidental ingestion or poisoning, particularly with unsupervised young children, occupational exposure, or the use of contami-

nated cannabinoids [12, 25–27]. Long-acting VKAs have a delayed onset of action, with their impact on coagulation tests often taking a few days to become apparent and a very long elimination half-life of 15 days to several months [28, 29]. People with altered VK pharmacogenomics that increase sensitivity to VKAs (e.g., polymorphisms in *VKORC1*, the gene encoding *VKORC1*, or in *CYP2C9*, the gene encoding cytochrome P450 2C9) [30] could be at greater risk of bleeding when exposed to long-acting VKAs. In contrast, those with polymorphisms that increase resistance (such as polymorphisms within *CYP4F2*, the gene encoding cytochrome 4F2) may be at lower risk [30]. Treating coagulopathies caused by long-acting VKAs involves administering intravenous VK before transitioning to oral VK, often at very high doses (e.g., 15–600 mg/day or 3.5–7 mg/kg/day, divided into four doses/day) for 1.5–6 months [28, 29, 31], to correct the VKD coagulopathy until the effects of the long-acting VKA resolve. Several reference laboratories offer brodifacoum assays, which assist in determining the treatment duration [32] without interrupting VK treatment.

Although poisoning from long-acting VKAs is rare, VKD arising from other causes is not. A high suspicion of VKD should be maintained when evaluating neonates and children with bleeding, as well as patients of any age with an acquired coagulopathy, considering that VKD is prevalent among patients with many acute and chronic illnesses. For example, the estimated prevalence of VKD in children was ~54% for those with ulcerative colitis and ~44% for those with Crohn's disease [33]. For children and young adults with severe neurological impairments, the estimated prevalence of VKD was ~43% (with ~78% presenting with a bleeding tendency) [34], which likely reflects the effects of antiepileptic medications, gastroesophageal reflux, reduced dietary VK intake, and recent antibiotic use. VKD was also present in 20–40% of critically ill patients, although it is not predictive of altered mortality in hospitalized patients [6, 35]. Subclinical VKD is also common in chronic kidney disease (CKD), with an estimated prevalence of 29–93%, based on laboratory findings for adults on hemodialysis [36]. Previous data showed that subclinical VKD was present in ~36% of older adults admitted with hip fractures, which increased to ~64% during hospitalization [37]. Subclinical VKD is also common among adults with advanced cancer who are receiving palliative care (~22–78%) [38].

VK supplementation improved mild INR prolongations in patients in intensive care units (ICUs) [39]. However, much larger studies would be needed to determine whether it can reduce bleeding risks in this population. Although VKD has been associated with poorer outcomes in CKD, VK supplementation did not

improve these outcomes [40, 41]. Treating critically ill patients with liver disease with VK supplements improved their INR by a median value of 0.63 without a measurable clinical benefit, such as reduced bleeding [42]. A recent report demonstrated improved outcomes for patients with chronic liver failure who received VK supplementation [43]; however, this finding would need to be corroborated in prospective studies. VK supplementation is a reasonable consideration for patients with disseminated intravascular coagulation, purpura fulminans, and/or those at increased risk for VKDB or microvascular thrombosis, such as that resulting from ischemic hepatitis, suspected heparin-induced thrombocytopenia, or cancer-related thrombosis [44, 45].

In infants and children with suspected VKDB whose coagulopathy does not completely reverse within several days of VK treatment, the possibility of a familial VK-dependent clotting factor deficiency (VKCFD) merits consideration [7]. VKCFD results from autosomal recessive pathogenic variations of *GGCX* or *VKORC1*, two enzymes involved in the γ -carboxylation of VK-dependent proteins (Fig. 1) [7, 46]. These disorders are more frequent in families with consanguineous parentage [7, 46].

LABORATORY DIAGNOSIS OF VKD

VKD diagnosis requires clinical suspicion and an evaluation of coagulation-test findings for abnormalities that could reflect VKD [6], as summarized in Table 2.

Diagnostic tests for VKD have been challenging to design because VK circulates at low levels and in different forms, with differing half-lives and plasma levels influenced by lipid profiles (reviewed in [6]). Different forms of VK can be detected using high-performance liquid chromatography [47]; however, such testing does not help attribute a coagulopathy to VKD. A diagnosis of coagulopathy from VKD is often based on coagulation-test findings, including a significant correction or reversal of abnormalities after VK treatment [6]. Sometimes, VKD diagnosis is made by simply confirming that a prolonged INR/PT fully corrects into the normal range within the expected time range after VK administration [6]. In infants and young children with suspected VKD, significant coagulopathy correction should occur within 2 hrs of administering 1 mg VK via intravenous (IV) or intramuscular (IM) injection [15]. In adults administered emergency treatment for severe bleeding with 10 mg VK (IV route) to reverse the effects of warfarin, coagulopathy typically improves within a few hours. A second 10 mg IV dose is recommended 12 hrs later to prevent rebound INR increases, which can occur ~12 hrs after

Table 2. Laboratory findings related to vitamin K deficiency

Tests	Typical findings
INR/PT	Increased, because of low levels of FII, FVII (shortest half-life), and FX
APTT	Increased, reflecting low levels of FII, FIX, and FX
Fibrinogen	Normal (or increased), unless a concomitant coagulopathy of another cause (e.g., liver disease) is involved
Factor levels	Decreased levels of FII, FVII, FIX, and FX; Other factor levels (including fibrinogen, FIIIE) are typically normal unless a concomitant coagulopathy of another cause (e.g., liver disease, congenital factor deficiencies) is involved, which is relatively common
Factor II activity levels estimated by PT (FII) vs. Ecarin prothrombin activator (FIIIE) reagents	Low FII/FIIIE (<0.86) because of reduced γ -carboxylation of VKDFs; Increased absolute difference between the levels of FIIIE minus FII (≥ 0.045 U/mL); Note: FIIIE levels <0.5 U/mL suggest concomitant liver disease
Anticoagulant proteins	Decreased levels of proteins C, S, and Z (rarely measured in suspected VKD)
Platelets	Normal
D-dimer	Usually normal but may be elevated in patients with large hematomas or concomitant abnormalities, such as liver disease or acute illness
PIVKA-II	Low, typically corrected within 4 days of VK treatment
Molecular investigations	Inherited VK-dependent clotting factor deficiencies result from variations in either <i>VKORC1</i> (encoding the 2,3-epoxide reductase complex that reduces vitamin K) or <i>GGCX</i> (encoding the γ -glutamyl carboxylase, which is involved in post-translational modification of vitamin K-dependent proteins)

Abbreviations: APTT, activated partial thromboplastin time; FII, factor II; FIIIE, factor II measured by one-stage assays with Ecarin prothrombin activator; FII/FIIIE, factor II/FIIIE ratio; FIX, factor IX; FX, factor X; GGCX, γ -glutamyl carboxylase; INR/PT, international normalized ratio/prothrombin time; PIVKA-II, proteins induced by vitamin K absence II; PT, prothrombin time; VK, vitamin K; VKDF, vitamin K-dependent factor; VKORC1, vitamin K epoxide reductase complex subunit 1.

the first dose [48].

Due to the shorter half-life of FVII than that of other vitamin K-dependent factors (VKDFs), VKD can present with an isolated, prolonged INR/PT [6], particularly when testing is done before reductions in the levels of FII, FIX, and FX prolong the activated partial thromboplastin time (APTT). Unless concerns regarding thrombosis or purpura fulminans exist, protein C (which has a short half-life similar to that of FVII) and protein S are not routinely evaluated in confirmed or suspected VKD. Fibrinogen and the levels of other non-VKDFs (i.e., FV, FVIII, FXI, and FXII) should remain within the normal range in VKD unless a mixed coagulopathy occurs (for example, from coexisting liver disease, where VK replacement often leads to improvement) [6].

Ecarin reagents contain a metalloproteinase prothrombin activator from *Echis carinatus* venom that converts both normal FII and des- γ -carboxylated FII to meizothrombin, which is an active form of thrombin [49, 50]. As γ -carboxylation affects FII activity in chromogenic or clot-based, one-stage FII assays with PT (FII) but not with Ecarin prothrombin-activator reagents (FIIIE), some laboratories diagnose VKD by testing for discrepant FII and FIIIE levels. VKD typically results in higher FII levels measured by FIIIE than by FII assays with PT reagents [6, 51]. Initial findings obtained by performing chromogenic assays to quantify FII and FIIIE indicated that an FII: FIIIE ratio of <0.86 or an increased abso-

lute difference between FIIIE and FII levels of ≥ 0.045 U/mL were predictive of VKD [51]. A recent consecutive-case cohort study, which included many patients with active liver disease (83%), was conducted to assess VKD using one-stage, clot-based FII and FIIIE assays and evaluate significant coagulopathy correction after VK treatment (defined as an INR correction/improvement of ≥ 0.5 after VK) [6]. The results of that study showed that FII: FIIIE ratios of <0.86 had 47.7% sensitivity and 90.2% specificity for VKD, and absolute differences of >0.04 U/mL had 69.3% sensitivity and 67.1% specificity for VKD [6]. As the cohort evaluated had high morbidity and mortality and included many people with liver disease, the estimated sensitivity and specificity may have been underestimated [6]. Nonetheless, among patients with FII: FIIIE ratios or absolute differences indicative of VKD, significant or complete coagulopathy correction was seen in 19/19 patients without liver disease and in 34/53 (64%) patients with active liver disease [6]. When evaluating FII: FIIIE discrepancies, it is also helpful to consider whether the FIIIE levels are <0.5 U/mL, which often indicates active liver disease [6]. Although low FII: FIIIE ratios are more common among patients with coagulopathies exclusive to VKDFs, some patients with VKD have mixed coagulopathies with low levels of non-VKDFs from concomitant liver disease (e.g., most commonly associated with low FVII, FXI, and/or FXII levels) [6].

Few laboratories offer assays for detecting PIVKA proteins (typically PIVKA-II) [6, 52]. PIVKA-II has been used to assess the presence of subclinical VKD [11]; however, the test is more commonly used for biomarker assessment of patients with hepatocellular carcinoma [53].

Incomplete correction of coagulopathy after VK treatment should raise suspicions regarding other causes, including active liver disease, congenital factor deficiencies [6] (e.g., mild inherited FVII deficiency), consumptive processes (such as disseminated intravascular coagulopathy), rodenticide poisoning that requires prolonged VK treatment for coagulopathy correction [12, 25, 31], and, on rare occasions, congenital VKD caused by *GGCX* or *VKORC1* variations, which requires confirmation via molecular testing [7].

PREVENTION AND TREATMENT OF VKD

Table 3 summarizes recommendations for prophylaxis to pre-

vent VKDB and for treating VKD and VKDB. The utility of VK prophylaxis at birth for preventing fatal VKDB was first demonstrated in the 1940s by Lehmann [54]. The European Society of Pediatric Gastroenterology, Hepatology, and Nutrition and the North American Society for Pediatric Gastroenterology currently recommend that all infants receive either: 1 mg VK prophylaxis at birth via IM injection; three oral doses of 2 mg VK at birth, 4–6 days later, and 4–6 weeks later; or 1 mg VK orally each week for 3 months [15]. The Canadian guidelines are similar, with additional dosing recommendations for smaller or premature infants and a variation in the timing of the second and third oral doses (at 2–4 weeks, then at 6–8 weeks), should oral VK be administered [55]. Other guidelines recommend three oral doses during the first 4–8 weeks of life or a large oral dose at birth, followed by smaller daily doses over the first 13 weeks [1]. However, oral VK prophylaxis may not prevent VKDB, particularly for infants at risk for malabsorption (e.g., because of biliary atresia) [11]. Spontaneous hemopericardium with cardiac tampon-

Table 3. Suggested management of bleeding from vitamin K deficiency by age

Age of onset	Prevention of VKD and bleeding	Treatment of bleeding from VKD
First 24 hrs up to 6 months of life	Term infants: 1 mg (or 0.5 mg when weighing <1,500 g) IM VK administration at birth; OR treatment (less effective) with 2 mg VK on day 1, day 4 (or 2–4 weeks), and week 4 (or 6–8 weeks) of life Premature infants: IM administration of 0.3 mg VK at birth Additionally: Mothers receiving VK-interfering medications (excluding warfarin) treated with 15–30 mg VK daily × 2–4 weeks before delivery (insufficient evidence supporting this treatment); OR treatment of their infants at birth with VK 0.5–1.0 mg IV (repeat dosing often required) Typical contents: Breast milk: 2.5 mg/L Formula: 24–175 mg/L	VK treatment: 240–300 µg/kg OR treatment with 1–2 mg IV once (increase to 3 mg when maternal VK antagonist is used); Administer VK SC when IV administration is not possible (avoid IM because of increased bleeding risk) For severe bleeding or bleeding in pre-term infants with hepatic immaturity: administer treatment with 10–15 mL/kg FFP or 50–100 units/kg PCC in addition to VK ICH: treat as described above and consider rFVIIa
Children >6 months	Recommended dietary intake: 6–12 months: 5–10 µg/day; 1–10 yrs: 15–30 µg/day; 11–17 yrs: 55–65 µg/day Chronic cholestasis: 2.5–5 mg daily or twice weekly	IV VK administration: 100 µg/kg, up to 10 mg For active bleeding, administer additional treatment (50 units/kg PCC or 15 mL/kg FFP)
Adults	Recommended dietary intake: Women: 90 µg/day; Men: 120 µg/day TPN: 150 µg VK daily	1–10 mg VK administered OR, SC, or IV (higher doses of IV VK to be considered for more severe coagulopathies), consider repeat dosing in 12 hrs If severe bleeding (e.g., ICH on a VKA): 10 mg VK IV (administered no faster than 1 mg/min) and consider 50 units/kg PCC and 1.2–4.8 mg rFVIIa IV Consult recent recommendations for treating bleeding from VKA and managing supratherapeutic INRs on a VKA. Note: VKD secondary to a long-acting VKA often requires treatment with higher than normal doses of VK IV (>10 mg daily) followed by OR therapy (with an equivalent dose) for several months, with adjustments based on responses/test findings

The Table summarizes information from [1, 11, 12, 15, 55, 65].

Abbreviations: FFP, fresh frozen plasma; ICH, intracranial hemorrhage; IM, intramuscular; INR, international normalized ratio; IV, intravenous; OR, oral; PCC, prothrombin complex concentrates; rFVIIa, recombinant activated factor VII; SC, subcutaneous; TPN, total parenteral nutrition; VK, vitamin K (for treatment purposes, phytonadione is a vitamin K1 treatment available in oral, SC, IM and IV forms); VKA, vitamin K antagonist (e.g., warfarin) therapy; VKD, vitamin K deficiency.

ade was reported in an 11-week-old infant with alpha-1 antitrypsin deficiency who had received three scheduled oral doses of VK [56]. The results of a systematic review showed that multiple oral doses of VK were non-inferior to a single IM dose in preventing VKDB [57]; however, higher rates of late VKDB were observed in those receiving oral VK prophylaxis [11, 15], possibly because of poor compliance.

Public health data suggest that parenteral VK treatment is more effective than oral VK treatment. For example, after Australia updated its national guidelines to recommend IM rather than oral VK prophylaxis in 1995, the rate of infantile VKDB fell from 2.46 per 100,000 births to 0.56 per 100,000 births [58]. Similarly, updated guidance from the American Academy of Pediatrics in 2003 recommending IM VK prophylaxis reduced the rate of classic and late VKDB to <2 per 100,000 infants [15]. Thus, IM VK prophylaxis is preferred for preventing VKDB in infants [11, 15]. Notably, no significant adverse effects have been reported following IM VK treatment [55], and the potential association between childhood cancer and IM VK (reported in a single study) has not been verified in other studies [59]. Parental refusal of IM VK prophylaxis for newborns has been increasing in some regions [58]. Providing parents with information on the associated risks is important, particularly the increased risk of VKDB when oral VK is used [1, 55]. Mild adverse effects of IM VK include localized site irritation and a theoretical risk of infection at the injection site [15]. When IM VK is refused, an appropriate oral VK prophylaxis schedule should be advised. The IV route has not been extensively studied in infants and children; however, IV VK is recommended for treating VKDB [15].

Ensuring that newborns delivered at home receive VK prophylaxis is important, as these infants may be at risk for VKDB [17]. PIVKA-II levels in cord blood suggest that pre-term and full-term infants have comparable VK requirements [1]. Pre-term infants should receive 0.3 mg VK IM when weighing less than 1,000 g and 0.5 mg when weighing between 1,000–1,500 g [11]. Infants with higher birth weights should receive 1 mg VK IM [55].

Beyond the neonatal period, the recommended daily oral VK intake for infants, children, and adolescents is 5–10 µg VK/day, 15–30 µg/day, and 55–65 µg/day, respectively [11]. For adult males and females, the recommended oral VK intake is 120 and 90 µg/day, respectively [60]. Although breastmilk tends to be a poor source of VK, supplementing breastfeeding mothers with 5 mg VK/day improves infant PIVKA-II levels [11, 61]. Presently, insufficient evidence is available to support the use of pre-natal VK supplementation for mothers taking non-anticoagulant medications that affect VK metabolism [1, 15]. Although oral VK

is an alternative to IM VK administration for correcting PIVKA-II levels within the first week of life [15], parenteral treatment (IM injection) is recommended for infants of mothers on medications that interfere with VK metabolism and for treating patients unable to take or absorb oral VK [11, 15]. For infants and children with malabsorptive conditions, the recommended dosing may not be high enough to combat subclinical VKD and/or reduced bone density because of decreased osteocalcin carboxylation [11].

In adults, the bleeding risk associated with VKD is uncertain, although bleeding while on a VKA is a well-recognized problem [62]. For adults taking warfarin who present with intracranial hemorrhaging and need urgent reversal of VKD, 10 mg of intravenous VK should be administered as a slow infusion over 30 min (due to an estimated risk of anaphylaxis of 3/100,000 people). This dose should be administered at presentation and 12 hrs later, along with rapid reversal treatment with prothrombin complex concentrates [48, 63]. Poisoning with long-acting rodenticides requires very prolonged courses of VK at higher doses [25, 27, 29, 31, 64].

SUMMARY AND OPPORTUNITIES FOR IMPROVEMENT

VKD is the most common cause of acquired coagulopathy across all age groups. It can develop within days of reduced VK intake, prenatally in fetuses of mothers with VKD, or shortly after birth when VK prophylaxis is not administered. Diagnosing VKD remains challenging because of the different forms and half-lives of VK and the limited sensitivity of coagulation assays, which may be less sensitive than PIVKA and VK1 assays. Empirical VK treatment and assessment of coagulopathy correction remain the most common diagnostic approaches. Assessing discrepancies in FII levels estimated by PT versus Ecarin reagents expedites VKD diagnosis and can help detect VKD in patients with coexisting coagulopathies, including those that affect both VKDF and non-VKDFs (such as in liver disease). Increasing physician awareness is crucial, as VKD accounts for at least 25% of cases referred for coagulopathy evaluation and represents the most common vitamin deficiency encountered in clinical practice.

Important tips

1. Serious bleeding in infants and children is likely to result from VKD unless proven otherwise, especially when VK prophylaxis was not administered at birth. We recommend that physi-

cians treat first and assess the laboratory results later.

2. Be suspicious of VKD in older children and adults. Many individuals have insufficient dietary VK intake. Furthermore, VKD is relatively prevalent among individuals who are acutely or chronically ill.

3. VKD can occur in patients with or without other coagulopathies. Consider empirical treatment for VKD followed by assessment for coagulopathy correction or improvement or conduct specific investigations for VKD when such testing is available locally.

ACKNOWLEDGEMENTS

None.

AUTHOR CONTRIBUTIONS

Mathews N drafted the manuscript, with contributions from Hayward CPM. Hayward CPM prepared Fig. 1.

CONFLICTS OF INTEREST

Mathews N has no potential conflicts of interest to disclose. Hayward CPM has received honoraria from Stago Canada, Ltd. and Werfen, as well as royalties from UpToDate.

RESEARCH FUNDING

Hayward CPM received research funds from McMaster University. Mathews N received fellowship funding from the Department of Hematology/Oncology, Centre Hospitalier Universitaire Sainte-Justine, Montréal, Québec, Canada.

REFERENCES

1. Araki S and Shirahata A. Vitamin K deficiency bleeding in infancy. *Nutrients* 2020;12:780.
2. Eden RE, Daley SF, Coviello JM. Vitamin K deficiency. Treasure Island: StatPearls Publishing, 2024.
3. Mladěnka P, Macáková K, Kujovská Krčmová L, Javorská L, Mrštná K, Carazo A et al. Vitamin K – sources, physiological role, kinetics, deficiency, detection, therapeutic use, and toxicity. *Nutr Rev* 2022;80:677–98.
4. Furie B and Furie BC. Molecular basis of vitamin K-dependent gamma-carboxylation. *Blood* 1990;75:1753–62.
5. Hao Z, Jin DY, Stafford DW, Tie JK. Vitamin K-dependent carboxylation of coagulation factors: insights from a cell-based functional study. *Hematologica* 2020;105:2164–73.
6. Bourguignon A, Mathews N, Tasneem S, Douketis J, Hayward CPM. Rapid diagnosis of coagulopathies from vitamin K deficiency in a consecutive case cohort evaluated by comparative assessment of factor II by 1-stage assays with prothrombin time vs Ecarin reagents. *J Thromb Haemost* 2024;22:3059–69.
7. Mathews N, Tasneem S, Hayward CPM. Rare inherited coagulation and fibrinolytic defects that challenge diagnostic laboratories. *Int J Lab Hematol* 2023;45(S2):30–43.
8. Shearer MJ and Newman P. Recent trends in the metabolism and cell biology of vitamin K with special reference to vitamin K cycling and MK-4 biosynthesis. *J Lipid Res* 2014;55:345–62.
9. Iioka H, Moriyama IS, Morimoto K, Akada S, Hisanaga H, Ishihara Y, et al. Pharmacokinetics of vitamin K in mothers and children in the perinatal period: transplacental transport of vitamin K2 (MK-4). *Asia Oceania J Obstet Gynaecol* 1991;17:97–100.
10. Halder M, Petsophonsakul P, Akbulut AC, Pavlic A, Bohan F, Anderson E, et al. Vitamin K: double bonds beyond coagulation insights into differences between vitamin K1 and K2 in health and disease. *Int J Mol Sci* 2019;20:896.
11. Van Winckel M, De Bruyne R, Van De Velde S, Van Biervliet S. Vitamin K, an update for the paediatrician. *Eur J Pediatr* 2009;168:127–34.
12. Watt BE, Proudfoot AT, Bradberry SM, Vale JA. Anticoagulant rodenticides. *Toxicol Rev* 2005;24:259–69.
13. McNinch A. Vitamin K deficiency bleeding: early history and recent trends in the United Kingdom. *Early Hum Dev* 2010;86(S1):63–5.
14. Dam H. The antihemorrhagic vitamin of the chick. *Biochem J* 1935;29:1273–85.
15. Mihatsch WA, Braegger C, Bronsky J, Campoy C, Domellöf M, Fewtrell M, et al. Prevention of vitamin K deficiency bleeding in newborn infants: a position paper by the ESPGHAN Committee on Nutrition. *J Pediatr Gastroenterol Nutr* 2016;63:123–9.
16. Pichler E and Pichler L. The neonatal coagulation system and the vitamin K deficiency bleeding – a mini review. *Wien Med Wochenschr* 2008;158:385–95.
17. Volpe JJ. Vitamin K deficient bleeding and COVID-19: how are they related? *J Neonatal Perinat Med* 2021;14:313–6.
18. Yasuda K, Obata S, Kumagai E, Iijima T, Yamamoto M, Miyagi E, et al. Intrauterine fetal death due to fetal intracranial hemorrhage associated with maternal vitamin K deficiency: a case report. *Case Rep Womens Health* 2023;40:e00566.
19. Sotodate G, Matsumoto A, Konishi Y, Toya Y, Endo M, Oyama K. Fetal intracranial hemorrhage due to maternal subclinical vitamin K deficiency associated with long-term eating disorder. *J Obstet Gynaecol Res* 2019;45:461–5.
20. Bechensteen BT, Sithiravel C, Strøm-Roum EM, Ruud HK, Kravdal G, Winther JA, et al. Post-bariatric pregnancy is associated with vitamin K1 deficiency, a case control study. *BMC Pregnancy Childbirth* 2024;24:229.
21. Eerdekens A, Debeer A, Van Hoey G, De Borger C, Sachar V, Guelinckx I, et al. Maternal bariatric surgery: adverse outcomes in neonates. *Eur J Pediatr* 2010;169:191–6.
22. Jans G, Guelinckx I, Voets W, Galjaard S, Van Haard PMM, Vansant GM, et al. Vitamin K1 monitoring in pregnancies after bariatric surgery: a prospective cohort study. *Surg Obes Relat Dis* 2014;10:885–90.
23. Van Mieghem T, Van Schoubroeck D, Depiere M, Debeer A, Hanssens M. Fetal cerebral hemorrhage caused by vitamin K deficiency after complicated bariatric surgery. *Obstet Gynecol* 2008;112:434–6.
24. Al-Zuhairi SH. Late vitamin K deficiency bleeding in infants: five-year prospective study. *J Pediatr (Rio J)* 2021;97:514–9.
25. Wright E, Hafner JW, Podolej G, Feinstein DL, van Breemen R, Rubinstein I, et al. Severe vitamin K-dependent coagulopathy from rodenticide-contaminated synthetic cannabinoids: emergency department presentations. *West J Emerg Med* 2021;22:1014–9.

26. Moritz E, Austin C, Wahl M, DesLauriers C, Navon L, Walblay K, et al. Notes from the field: outbreak of severe illness linked to the vitamin K antagonist brodifacoum and use of synthetic cannabinoids – Illinois, March–April 2018. *MMWR Morb Mortal Wkly Rep* 2018;67:607–8.
27. Arepally GM and Ortel TL. Bad weed: synthetic cannabinoid-associated coagulopathy. *Blood* 2019;133:902–5.
28. Hollinger BR and Pastoor TP. Case management and plasma half-life in a case of brodifacoum poisoning. *Arch Intern Med* 1993;153:1925–8.
29. Weitzel JN, Sadowski JA, Furie BC, Moroose R, Kim H, Mount ME, et al. Surreptitious ingestion of a long-acting vitamin K antagonist/rodenticide, brodifacoum: clinical and metabolic studies of three cases. *Blood* 1990;76:2555–9.
30. Moreau C, Bajolle F, Siguret V, Lasne D, Golmard JL, Elie C, et al. Vitamin K antagonists in children with heart disease: height and *VKORC1* genotype are the main determinants of the warfarin dose requirement. *Blood* 2012;119:861–7.
31. Katona B and Wason S. Superwarfarin poisoning. *J Emerg Med* 1989;7: 627–31.
32. Yip L, Stanton NV, Middleberg RA. Vitamin K₁ treatment duration in patients with brodifacoum poisoning. *N Engl J Med* 2020;382:1764–5.
33. Nowak JK, Grzybowska-Chlebowczyk U, Landowski P, Szaflarska-Poplawska A, Klineciewicz B, Adamczak D, et al. Prevalence and correlates of vitamin K deficiency in children with inflammatory bowel disease. *Sci Rep* 2014;4:4768.
34. Yoshikawa H, Yamazaki S, Watanabe T, Abe T. Vitamin K deficiency in severely disabled children. *J Child Neurol* 2003;18:93–7.
35. Paulus MC, Drent M, Kouw IWK, Balvers MGJ, Bast A, van Zanten ARH. Vitamin K: a potential missing link in critical illness—a scoping review. *Crit Care* 2024;28:212.
36. Voong K, Harrington D, Goldsmith D. Vitamin K status in chronic kidney disease: a report of a study and a mini-review. *Int Urol Nephrol* 2013; 45:1339–44.
37. Bultynck C, Munim N, Harrington DJ, Judd L, Ataklte F, Shah Z, et al. Prevalence of vitamin K deficiency in older people with hip fracture. *Acta Clin Belg* 2020;75:136–40.
38. Harrington DJ, Western H, Seton-Jones C, Rangarajan S, Beynon T, Shearer MJ. A study of the prevalence of vitamin K deficiency in patients with cancer referred to a hospital palliative care team and its association with abnormal haemostasis. *J Clin Pathol* 2008;61:537–40.
39. Dahlberg S, Schött U, Kander T. The effect of vitamin K on prothrombin time in critically ill patients: an observational registry study. *J Intensive Care* 2021;9:11.
40. Bellone F, Cinquegrani M, Nicotera R, Carullo N, Casarella A, Presta P, et al. Role of vitamin K in chronic kidney disease: a focus on bone and cardiovascular health. *Int J Mol Sci* 2022;23:5282.
41. Andrian T, Stefan A, Nistor I, Covic A. Vitamin K supplementation impact in dialysis patients: a systematic review and meta-analysis of randomized trials. *Clin Kidney J* 2023;16:2738–49.
42. Al Sulaiman K, Al Mutairi M, Al Harbi O, Al Duraihim A, Aldosary S, Al Khalil H, et al. Appropriateness of using vitamin K for the correction of INR elevation secondary to hepatic disease in critically ill patients: an observational study. *Clin Appl Thromb Hemost* 2021;27:10760296211050923.
43. Xiong Z, Liu Y, Chang T, Xu X, Huo S, Deng H, et al. Effect of vitamin K1 on survival of patients with chronic liver failure: a retrospective cohort study. *Medicine (Baltimore)* 2020;99:e19619.
44. Warkentin TE. Microvascular thrombosis and ischaemic limb losses in critically ill patients. *Haemostaseologie* 2019;39:6–19.
45. Levi M and Scully M. How I treat disseminated intravascular coagulation. *Blood* 2018;131:845–54.
46. Napolitano M, Mariani G, Lapecorella M. Hereditary combined deficiency of the vitamin K-dependent clotting factors. *Orphanet J Rare Dis* 2010;5:21.
47. Kamao M, Sahara Y, Tsugawa N, Okano T. Determination of plasma vitamin K by high-performance liquid chromatography with fluorescence detection using vitamin K analogs as internal standards. *J Chromatogr B Analyt Technol Biomed Life Sci* 2005;816:41–8.
48. Christensen H, Cordonnier C, Körv J, Lal A, Ovesen C, Purruccer JC, et al. European Stroke Organisation guideline on reversal of oral anticoagulants in acute intracerebral haemorrhage. *Eur Stroke J* 2019;4:294–306.
49. Mohammadi N, Bandehpour M, Sotoodehnejadnematalahi F, Kazemi B. Prokaryotic expression, evaluation, and prediction of the structure and function of the ecarin metalloproteinase domain. *Proteins* 2022;90: 802–9.
50. Stevens WK, Côté HF, MacGillivray RT, Nesheim ME. Calcium ion modulation of meizothrombin autolysis at Arg55–Asp56 and catalytic activity. *J Biol Chem* 1996;271:8062–7.
51. Bertina RM, van der Marel-van Nieuwkoop W, Dubbeldam J, Boekhout-Mussert RJ, Veltkamp JJ. New method for the rapid detection of vitamin K deficiency. *Clin Chim Acta* 1980;105:93–8.
52. Widdershoven J, van Munster P, De Abreu R, Bosman H, van Lith T, van der Putten-van Meyel M, et al. Four methods compared for measuring des-carboxy-prothrombin (PIVKA-II). *Clin Chem* 1987;33:2074–8.
53. Panettieri E, Campisi A, De Rose AM, Mele C, Giuliani F, Vauthey JN, et al. Emerging prognostic markers in patients undergoing liver resection for hepatocellular carcinoma: a narrative review. *Cancers (Basel)* 2024; 16:2183.
54. Lehmann J. Vitamin K as a prophylactic in 13,000 infants. *Lancet* 1944; 243:493–4.
55. Ng E and Leowy AD. Guidelines for vitamin K prophylaxis in newborns. *Paediatr Child Health* 2018;23:394–402.
56. Bauer C, Furthner D, Grohmann E, Tulzer G. Vitamin K deficiency-induced spontaneous haemopericardium and cardiac tamponade in an infant with alpha-1 antitrypsin deficiency: a case report. *Eur Heart J Case Rep* 2021;5:ytaa481.
57. Sankar MJ, Chandrasekaran A, Kumar P, Thukral A, Agarwal R, Paul VK. Vitamin K prophylaxis for prevention of vitamin K deficiency bleeding: a systematic review. *J Perinatol* 2016;36(S1):S29–35.
58. Zurynski Y, Grover CJ, Jalaludin B, Elliott EJ. Vitamin K deficiency bleeding in Australian infants 1993–2017: an Australian Paediatric Surveillance Unit study. *Arch Dis Child* 2020;105:433–8.
59. Roman E, Fear NT, Ansell P, Bull D, Draper G, McKinney P, et al. Vitamin K and childhood cancer: analysis of individual patient data from six case-control studies. *Br J Cancer* 2002;86:63–9.
60. Vitamin K. Fact sheet for health professionals. <https://ods.od.nih.gov/factsheets/VitaminK-HealthProfessional> (Updated on Mar 2021).
61. Greer FR, Marshall SP, Foley AL, Suttie JW. Improving the vitamin K status of breastfeeding infants with maternal vitamin K supplements. *Pediatrics* 1997;99:88–92.
62. Vazquez FJ, Gonzalez JP, LeGal G, Carrier M, Gándara E. Risk of major bleeding in patients receiving vitamin K antagonists or low doses of aspirin. A systematic review and meta-analysis. *Thromb Res* 2016;138:1–6.
63. Goodnough LT and Shander A. How I treat warfarin-associated coagulopathy in patients with intracerebral hemorrhage. *Blood* 2011;117: 6091–9.
64. Long J, Peng X, Luo Y, Sun Y, Lin G, Wang Y, et al. Treatment of a long-acting anticoagulant rodenticide poisoning cohort with vitamin K1 during the maintenance period. *Medicine (Baltimore)* 2016;95:e5461.
65. Shahrook S, Ota E, Hanada N, Sawada K, Mori R. Vitamin K supplementation during pregnancy for improving outcomes: a systematic review and meta-analysis. *Sci Rep* 2018;8:11459.